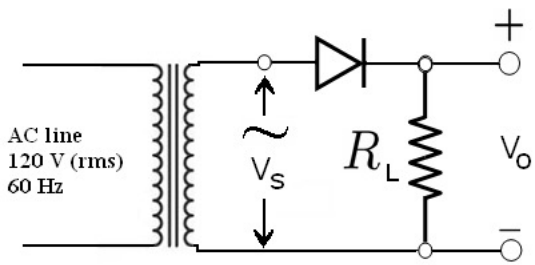
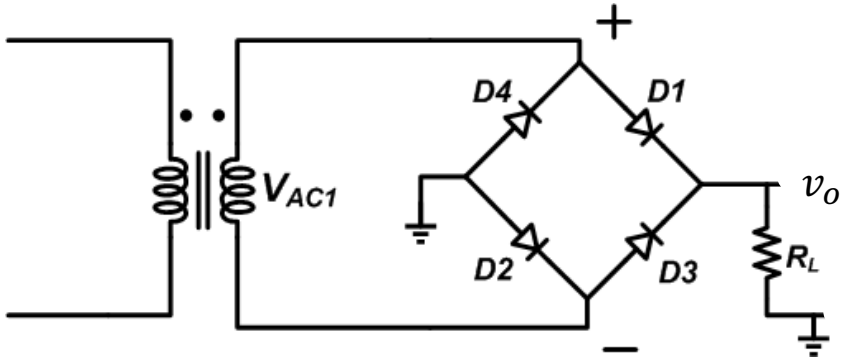
Analysis of Diode Circuits

- Rectifiers.
- Rectifiers with Peak detector.
- Regulation Using Zener Diodes.

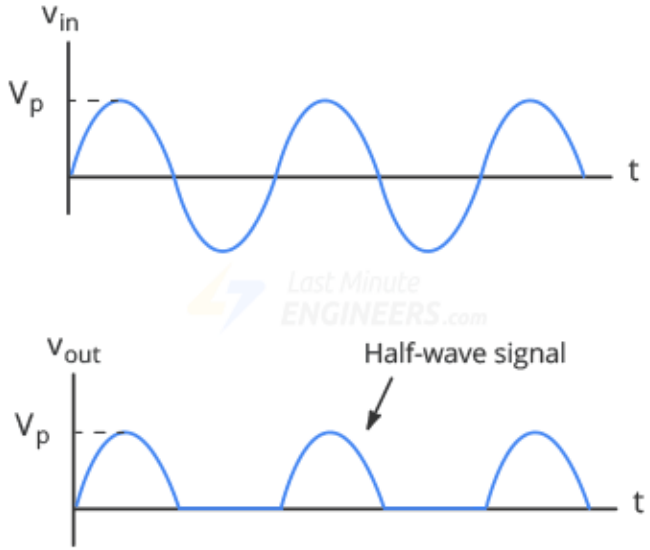
Rectifiers

Rectifiers convert the oscillating input signal to a one-directional signal

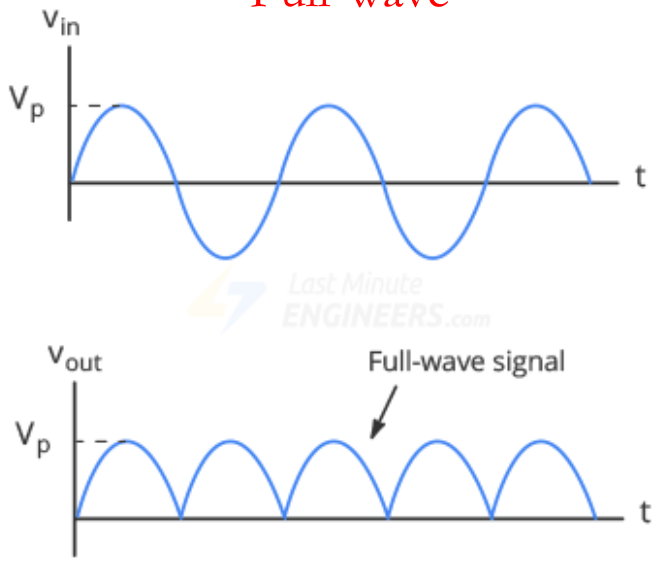




Half-wave

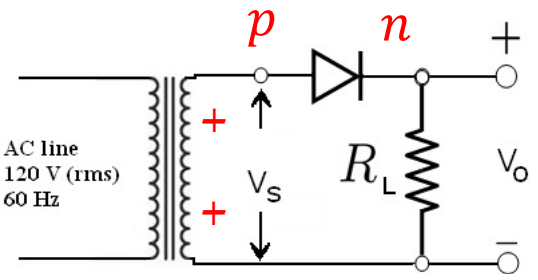


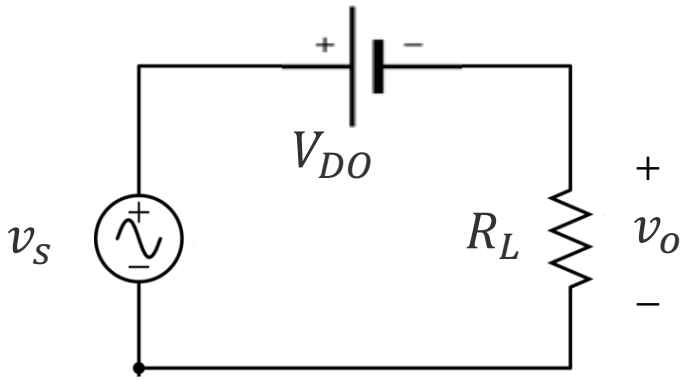
Full-wave



Half-wave Rectifiers

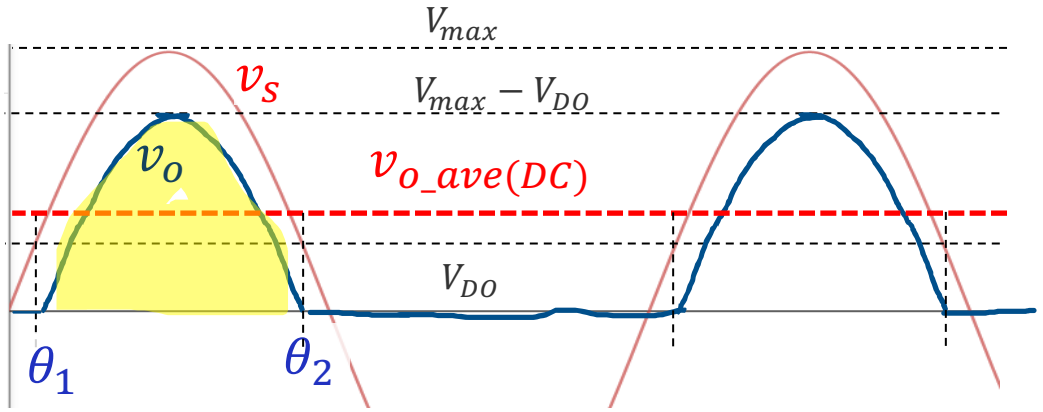
A half-wave rectifier utilizes only half of the input power.





$$v_s(t) = V_{max} \sin \omega t$$

Consider the constant-battery diode model $v_o(t) = v_s(t) - V_{DO}$



$$v_{o_ave} = \frac{1}{2\pi}\int_{\theta_1}^{\theta_2} v_o(\omega t)\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{2\pi}\int_{\theta_1}^{\theta_2} [v_s(\omega t) - V_{DO}]\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{2\pi}\int_{\theta_1}^{\theta_2} [V_{max} \sin \omega t - V_{DO}]\, d(\omega t)$$

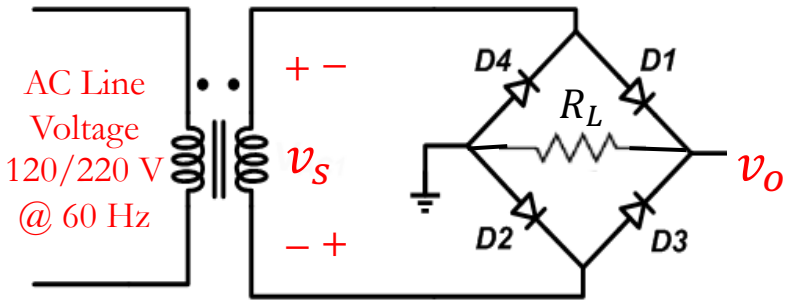
$$v_{o_ave} = \frac{-1}{2\pi}[V_{max}(\cos\theta_2 - \cos\theta_1) + V_{DO}(\theta_2 - \theta_1)]$$

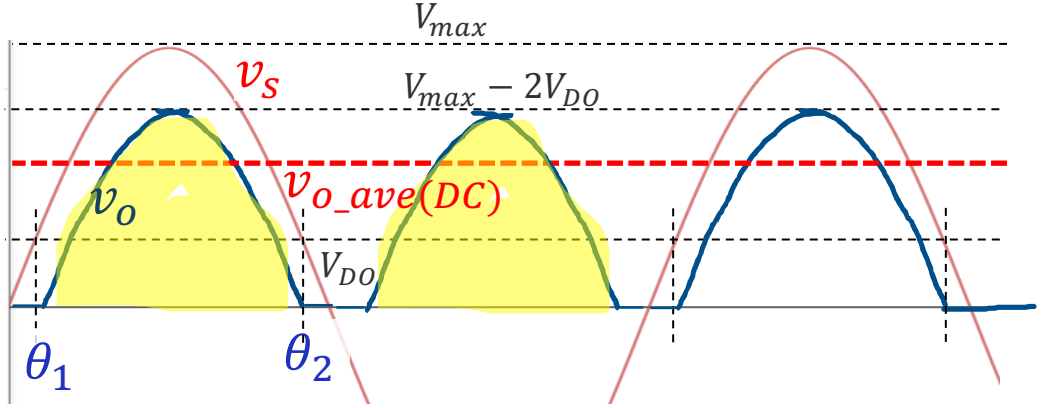
$PIV = V_{max}$ Peak Inverse Voltage

Peak conduction current $I_{max} = \dfrac{V_{max} - V_{DO}}{R}$

Full-wave Rectifiers

A full-wave rectifier utilizes both halves of the input power.





$$v_s(t) = V_{max} \sin \omega t$$

Consider the constant-battery diode model

$$v_o(t) = v_s(t) - 2V_{DO}$$

$$v_{o_ave} = \frac{2}{2\pi} \int_{\theta_1}^{\theta_2} v_o(\omega t)\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{\pi} \int_{\theta_1}^{\theta_2} [v_s(\omega t) - 2V_{DO}]\, d(\omega t)$$

$$v_{o_ave} = \frac{1}{\pi} \int_{\theta_1}^{\theta_2} [V_{max} \sin \omega t - 2V_{DO}]\, d(\omega t)$$

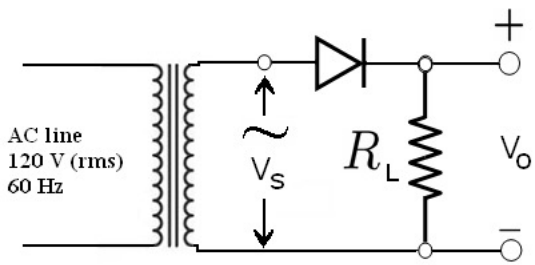
$$v_{o_ave} = \frac{-1}{\pi} [V_{max}(\cos \theta_2 - \cos \theta_1) + 2V_{DO}(\theta_2 - \theta_1)]$$

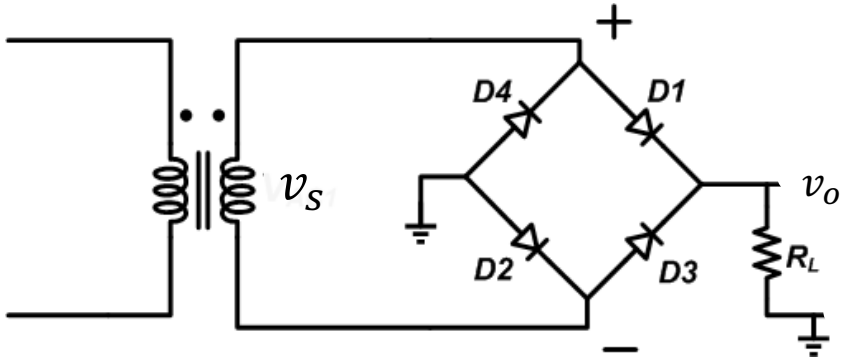
$PIV \approx V_{max}$ Peak Inverse Voltage

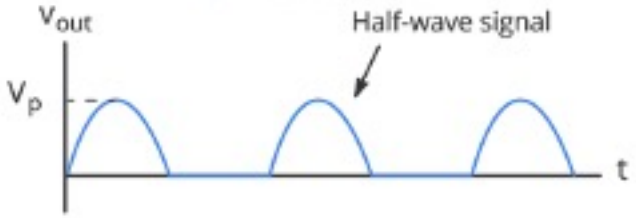
Peak conduction current $I_{max} = \dfrac{V_{max} - 2V_{DO}}{R}$

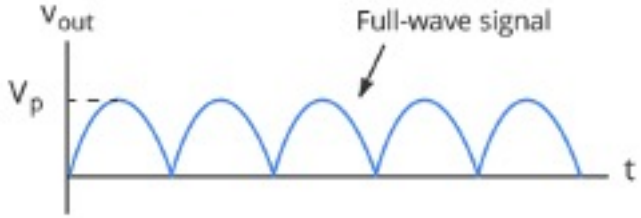
Rectifiers Transfer Function

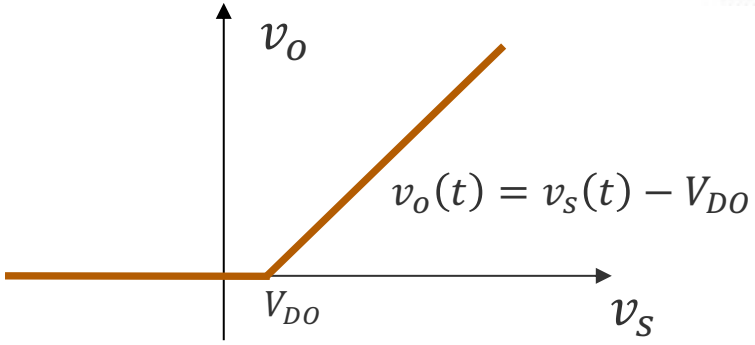
Rectifiers convert the oscillating input signal to a one-directional signal

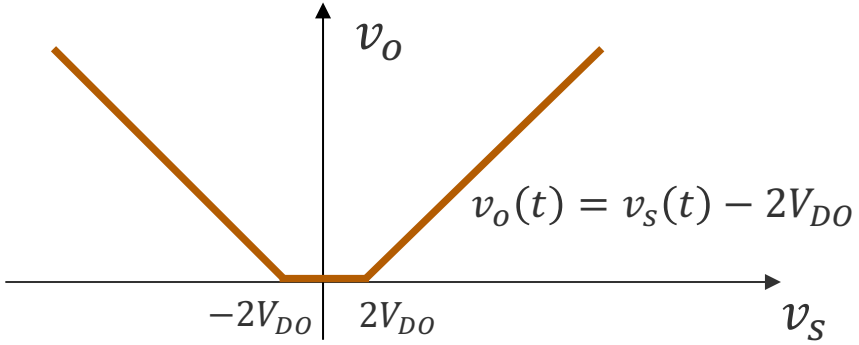






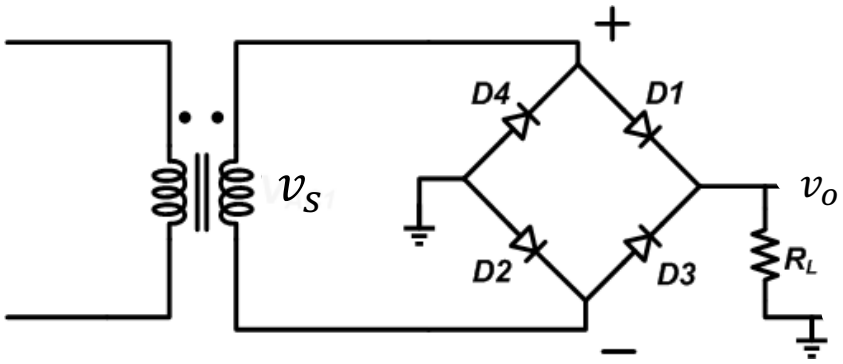






Example

A bridge rectifier is fed from a 120 V (rms) AC source at 60 Hz. The circuit uses a 10:1 transformer with a load of 1 kΩ. Analyze the circuit using the ideal diode model.



$$V_{max} = \frac{120}{10}\sqrt{2} = 12\sqrt{2} \text{ V}$$

$$v_{o_max} = V_{max} - 2V_{DO} = V_{max} = 12\sqrt{2} \text{ V}$$

$$v_{o_ave} = \frac{2V_{max}}{\pi} = 10.8 \text{ V}$$

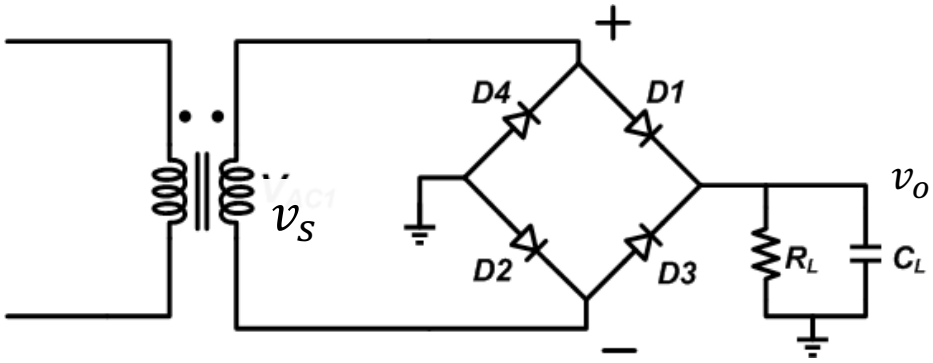
$$i_{D_ave} = \frac{v_{o_ave}}{R_L} = 10.8 \text{ mA}$$

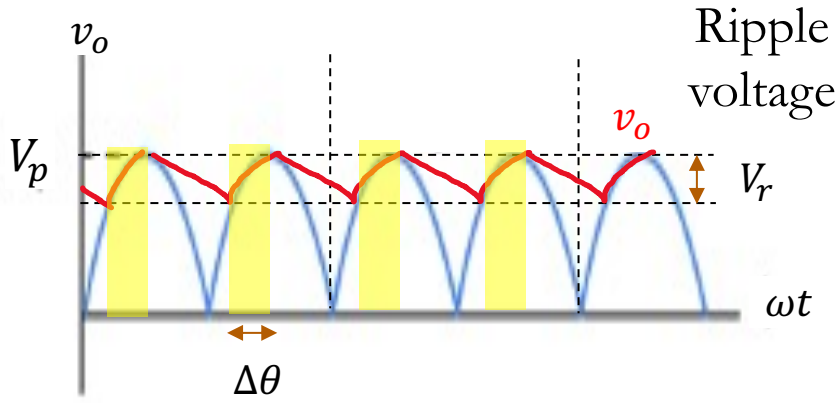
$$i_{D_max} = \frac{v_{o_max}}{R_L} = 17 \text{ mA}$$

$$PIV = 17 \text{ V}$$

Rectifier with Peak Detector

Consider a bridge rectifier with a shunt capacitor connected at the output (in parallel). Assume ideal diode model.





Conduction angle

$$v_{o_ave} \approx V_p - \frac{V_r}{2} \qquad V_r = \frac{V_p}{2fRC}$$

$$i_{D_ave} = I_L\left(1 + \pi\sqrt{\frac{V_p}{2V_r}}\right)$$

$$i_{D_max} = I_L\left(1 + 2\pi\sqrt{\frac{V_p}{2V_r}}\right)$$

$$I_L \approx V_p/R$$

$$\Delta\theta \approx \sqrt{2V_r/V_p}$$

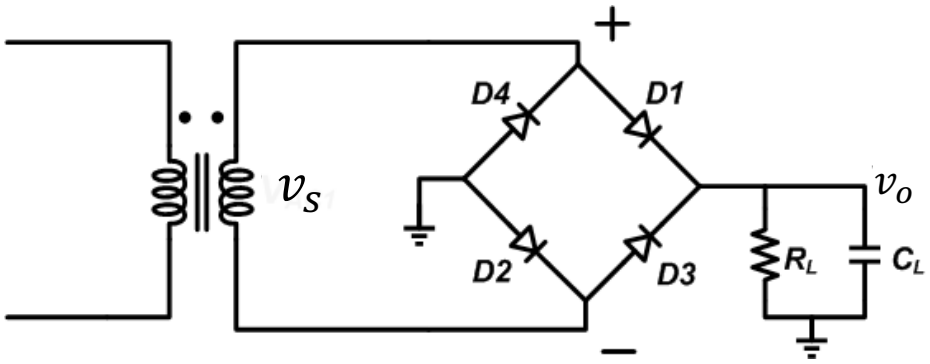
Notes: 1) For non-ideal diodes $V_p = V_{max} - 2V_{DO}$

2) For half-wave rectifier $V_p = V_{max} - V_{DO}$

$$V_r = \frac{V_p}{fRC}$$

Example

The output of the bridge rectifier in the previous example is connected to a peak detector (shunt capacitor). Design the detector such that the ripple voltage is no more than 1 V (p-p). Take $V_{DO} = 0.8$ V.



$$C = \frac{V_p - 2V_{DO}}{2fRV_r} = \frac{12\sqrt{2} - 2(0.8)}{2(60)(1\,k)(1)}$$

$$C = 128\ \mu F$$

$$v_{o_ave} = V_{max} - 2V_{DO} - \frac{V_r}{2} = 14.9\ V$$

$$i_{L_ave} = \frac{v_{o_ave}}{R_L} = 14.9\ \text{mA}$$

$$\Delta\theta \approx \sqrt{\frac{2V_r}{V_p - 2V_{DO}}} = 0.36\ \text{rad} \quad (20.7°)$$

$$i_{D_ave} = I_L\left(1 + \pi\sqrt{\frac{V_p - 2V_{DO}}{2V_r}}\right)$$

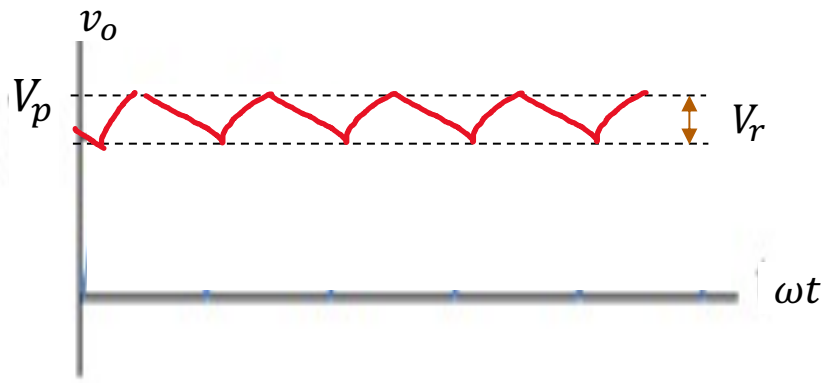
$$i_{D_ave} = 0.145\ A$$

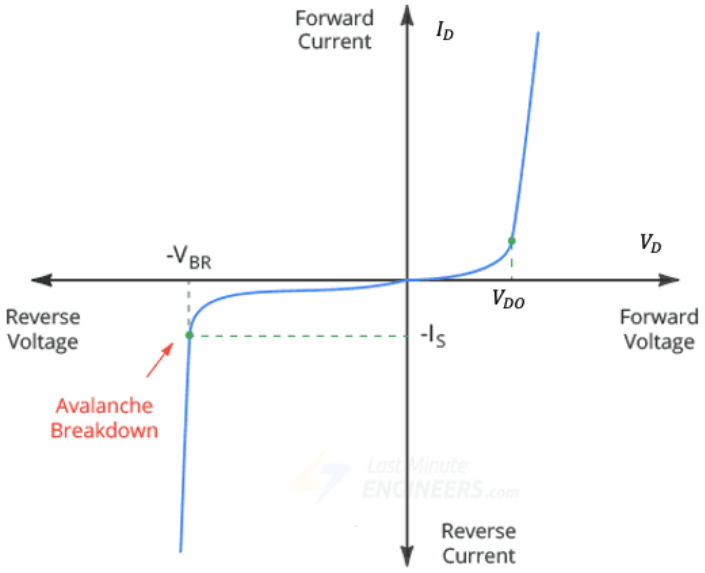
$$i_{D_max} = I_L\left(1 + 2\pi\sqrt{\frac{V_p - 2V_{DO}}{2V_r}}\right)$$

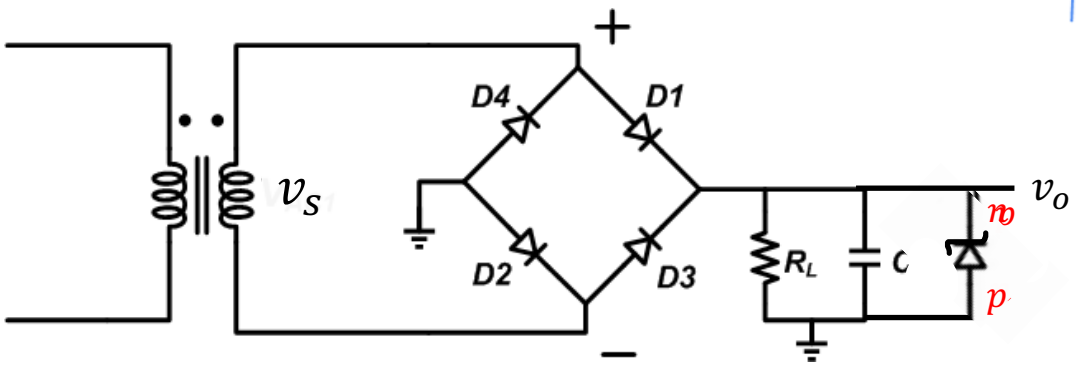
$$i_{D_max} = 0.274\ A$$

Regulation Using Zener Diodes

A diode connected at the peak detector output can remove the ripples in the output. How to connect the diode?







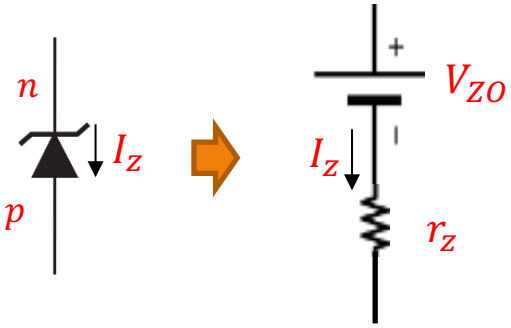
Zener Diode

D1N750 (3.6, 5.1 V)

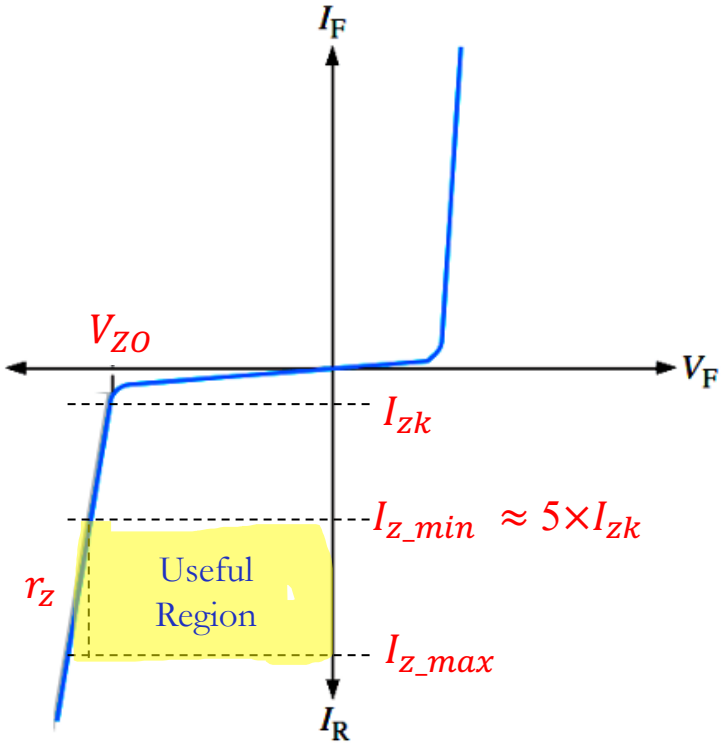
1N4732 (4.7 V)

Zener Diode Model

The Zener diode is modeled using the constant battery-R model. Three parameters need to be known (datasheet): V_{ZO}, r_z and I_{zk}.

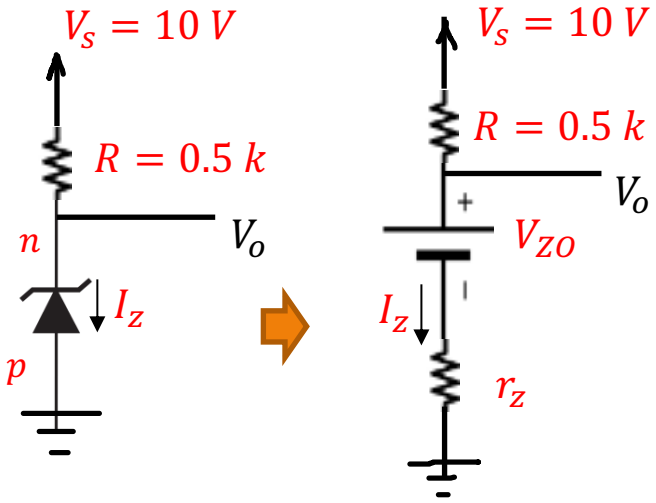


$$V_z = V_{ZO} + I_Z\, r_z$$



Example

1) The circuit below uses a Zener diode with the following parameters. The voltage source has no variations. $V_Z = 6.8\,V\ at\ 5\,mA, r_z = 20\,\Omega$ and $I_{zk} = 0.2\,mA$.



$$V_z = V_{ZO} + I_Z\, r_z \qquad V_{ZO} = V_Z - I_Z\, r_z$$

$$V_{ZO} = 6.8 - (5\,m)(20) = 6.7\,V$$

Is the diode in breakdown? YES

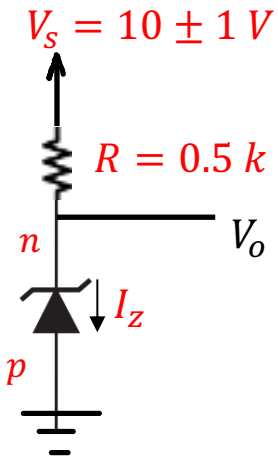
$$I_z = \frac{V_s - V_{ZO}}{R + r_z}$$

$$I_z = \frac{10 - 6.7}{500 + 20} = 6.35\,mA$$

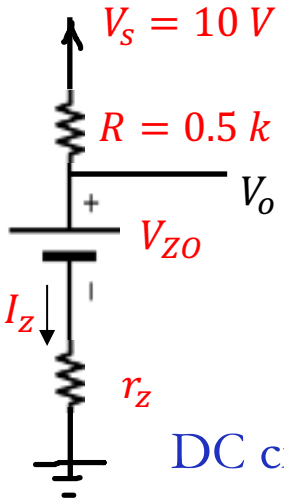
$$V_o = V_Z = V_{ZO} + I_Z\, r_z = 6.7 + (6.35\,m)(20) = 6.83\ \text{V}$$

Example

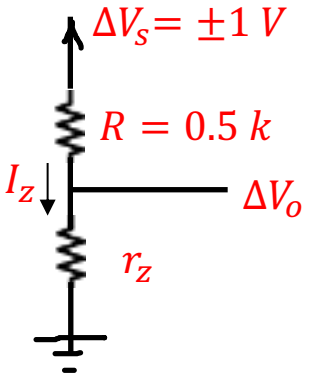
2) Now the source has a ±10% change. What will be the change in the output voltage?







DC circuit



AC circuit

±10% change in source

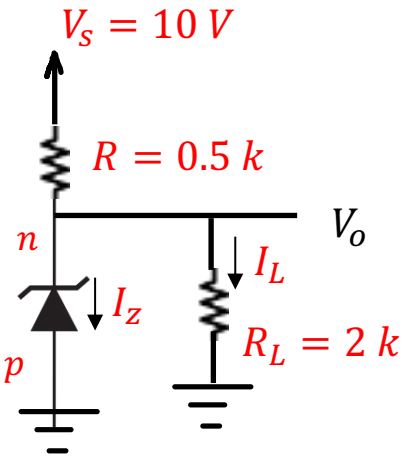
±0.6% change in output

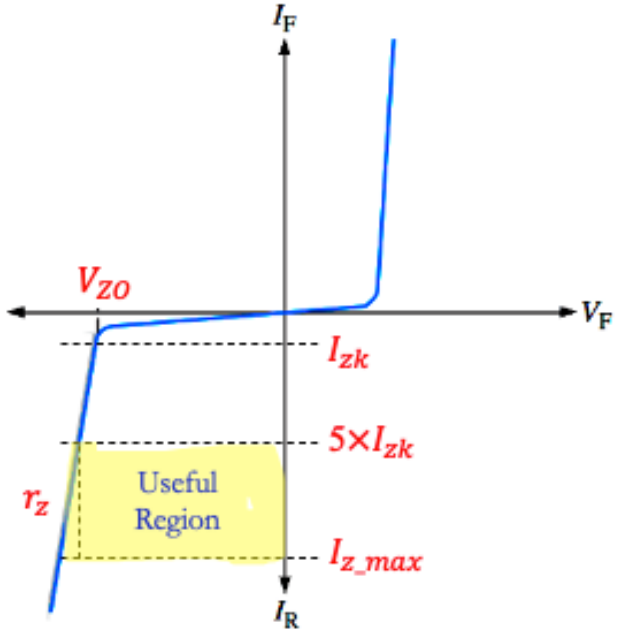
$$\Delta V_o = \Delta V_s \frac{r_z}{R + r_z}$$

$$\Delta V_o = \pm 1 \frac{20}{500 + 20} = \pm 38.5\ mV$$

Example

3) The source now is fixed at 10 V, but a load of 2 kΩ is connected. What is the change in output voltage?





$$I_L = \frac{V_o}{R_L} = \frac{6.83}{2\ k} = 3.42\ mA$$

$$I_z = 6.35 - 3.42 = 2.93\ mA$$

$$V_o = V_{ZO} + I_Z\, r_z = 6.7 + (2.93\ m)(20) = 6.76\ V$$

1% change in output

What if load is 0.5 k Ω?

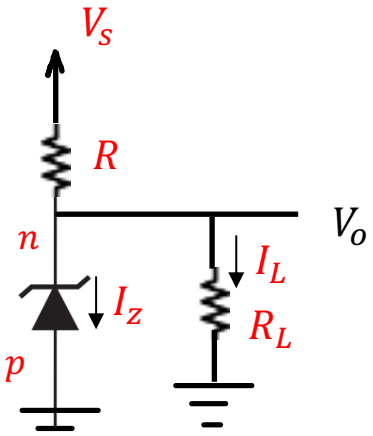
$$I_L = 6.83/2k = 13.66\ mA$$

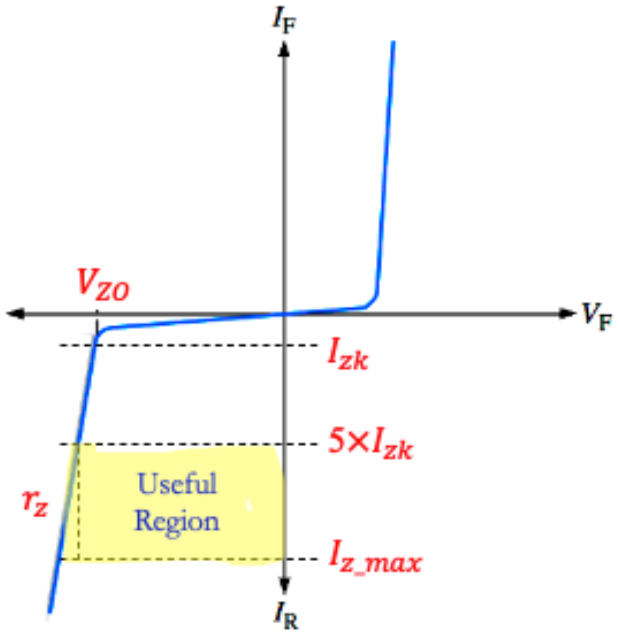
Diode is OFF

What is R_{L_min}?

Example

The resistance R should not be too high. Otherwise the diode will not get the minimum current for useful region?

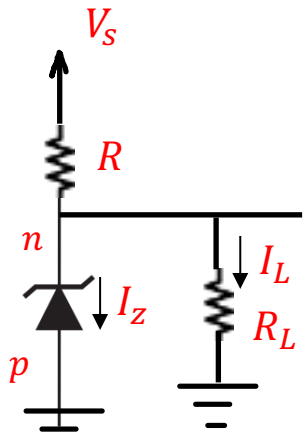




$$R_{max} = \frac{V_{s_min} - V_{ZO} - r_z\, I_{Z_min}}{I_{Z_min} + I_{L_max}}$$

Example

4) Design a shunt regulator to get an output voltage around 10 V. Use the 1N7470. The input source is expected to have 20±25% V and the load can take current up to 20 mA.

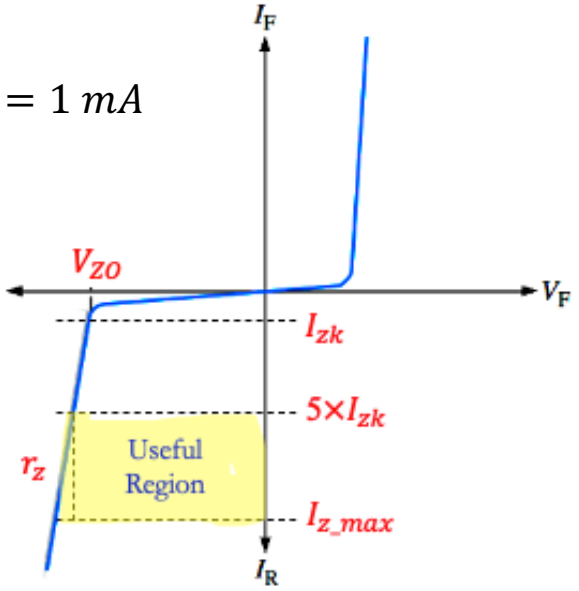


1N 7470: $V_Z = 10\ V\ at\ 25\ mA, r_z = 7\ \Omega$ and $I_{zk} = 1\ mA$

$$V_{ZO} = V_Z - I_Z\, r_z$$

$$V_{ZO} = 10 - (25\ m)(7) = 9.825\ V$$

$$R_{max} = \frac{V_{s_min} - V_{ZO} - r_z\, I_{Z_min}}{I_{Z_min} + I_{L_max}}$$



$$R_{max} = \frac{15 - 9.825 - (7)\ (5\ m)}{5\ m + 20\ m} = 205\ \Omega$$

$$R \leq 205\ \Omega$$

Q & A

Send me an email if
you have any
questions.